

Notes: Baron, Verner, and Xiong (QJE, 2021)

Banking Crises without Panics

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1 Big picture

Question. Are banking panics necessary for banking crises to generate deep recessions, or can large bank losses by themselves produce severe macroeconomic damage even when creditor runs are absent?

Core challenge. The classic panic-centered view of banking crises focuses on depositor or wholesale-funding runs, but many episodes of bank distress are “quiet” and therefore hard to detect with narrative chronologies alone. A panic-based chronology may miss crises in which banks are badly undercapitalized but creditor withdrawals are muted by guarantees, forbearance, or government intervention.

Main idea. The paper studies banking crises through *bank equity declines*. Equity holders are the first to absorb bank losses, so a large bank equity decline is a natural real-time signal that the banking system’s capacity to intermediate funds has been impaired.

Main data contribution. The authors construct a new historical data set of bank equity returns, nonfinancial equity returns, and narrative information on panics and widespread bank failures for 46 countries over 1870 to 2016, yielding nearly 2,500 country-years of annual data plus monthly data around crisis episodes.

Main empirical strategy. The paper estimates local projections of future real GDP growth and bank credit-to-GDP on bank equity returns, controlling for nonfinancial equity returns and standard macro lags. It then splits bank equity crash episodes into those with and without narrative panics.

Main baseline result. A bank equity crash — defined as an annual real bank equity return below -30% — predicts large and persistent macroeconomic damage. After three years, a 30% crash predicts about 3.4% lower real GDP and a 5.7 percentage point decline in bank credit-to-GDP.

Main asymmetry result. The predictive relation is strongly nonlinear. Large negative bank equity returns predict large output gaps and credit contractions, while positive returns do not symmetrically predict booms.

Main panic result. Panics amplify crises, but they are not necessary. Even *nonpanic* bank equity crashes predict sizable output losses and bank credit contraction. Conditional on narrative bank failures, nonpanic bank distress is about as severe as panic episodes.

Main timing result. Panics tend to come *after* bank equity has already fallen sharply. On average, the bank equity crash precedes the panic by about 7.5 months, suggesting that panics are more often an amplification of earlier bank losses than an independent first mover.

Main chronology contribution. Using bank equity losses, narrative panics, and widespread bank failures, the authors build the BVX Crisis List, a new banking crisis chronology that distinguishes bank equity crises from panic crises. The new chronology uncovers 27 additional crises omitted by prior narrative lists.

Economic takeaway. The defining feature of a banking crisis is a *bank capital crunch*. Panics matter, but the core damage often comes from bank losses and weak capitalization rather than from panic runs alone.

2 Data and measurement (key objects)

2.1 Annual bank and nonfinancial equity returns

For each country, the authors construct annual real total return indices for:

- **Bank equity:** price returns plus dividends for the banking sector.
- **Nonfinancial equity:** price returns plus dividends for a broad set of large nonfinancial firms.

The data set covers 46 advanced and emerging economies from 1870 to 2016. A large part of the series is newly assembled from hand-collected stock prices and dividend data from historical newspapers, statistical yearbooks, and archival sources. Existing sources such as Global Financial Data, Datastream, and earlier bank dividend data are also incorporated.

The bank index is meant to capture the broad domestic commercial banking sector, usually with especially strong coverage of the largest publicly traded banks in the country's financial center. The nonfinancial index is designed to represent broad corporate conditions while excluding sectors such as transportation, commodities, utilities, real estate companies, and colonial or foreign enterprises that are unusually exposed to external factors.

Interpretation. The nonfinancial equity index is the key control that lets the authors ask whether bank equity contains information *over and above* general stock market conditions.

2.2 Monthly returns and credit spreads

To study the timing of crises, the paper constructs monthly data in a three-year window around crisis episodes on:

- bank equity returns,
- nonfinancial equity returns,
- bank credit spreads, and
- nonfinancial corporate credit spreads.

The monthly panel covers 132 crisis episodes. Post-1980 monthly equity data mostly come from Datastream; earlier monthly data are assembled from newspapers, historical security manuals, Global Financial Data, and other archival sources.

Bank credit spreads are typically measured as short-term interbank lending rates minus Treasury bill yields. Corporate credit spreads are generally measured as corporate bond yields minus long-term Treasury yields.

Interpretation. The monthly data are not for estimating long-run macro effects; they are for establishing the *sequence* of events — whether bank equity falls before or after panics, spread spikes, and nonfinancial stock declines.

2.3 Macroeconomic variables

The paper studies two central macro outcomes:

- **Real GDP growth**, constructed from nominal or real GDP together with country CPIs.
- **Bank credit-to-GDP**, assembled mainly from the Jorda–Schularick–Taylor database, the BIS

long credit series, IMF statistical manuals, and League of Nations banking statistics.

These sources allow the authors to build long historical series for both output and bank intermediation.

Interpretation. Real GDP captures the broad macro consequence of banking distress; bank credit-to-GDP captures the specific *lending channel* through which bank losses are expected to matter.

2.4 Narrative crisis, panic, and bank-failure databases

The paper also constructs a narrative layer of crisis information.

Narrative Crises. A benchmark narrative crisis list is formed as the union of banking crisis dates from six influential papers, including Reinhart and Rogoff, Laeven and Valencia, and Schularick and Taylor.

Panics. A panic is defined broadly as a severe and sudden withdrawal of funding by bank creditors from a significant part of the banking system. This can include:

- classic depositor runs,
- sudden interbank market disruptions,
- severe foreign-currency funding outflows from the banking system.

Widespread bank failures. Widespread failure means either:

- the failure of a top-five bank by assets, or
- more than five bank failures above the normal rate.

Interpretation. Panics and failures are treated as *manifestations* of distress, not as the complete definition of crisis. That is exactly what allows the paper to distinguish bank distress with and without panics.

2.5 Main crisis objects

The paper uses several key crisis concepts.

Bank equity crash.

$$Crash_{i,t}^B \equiv \mathbf{1}\{r_{i,t}^B \leq -30\%\},$$

where $r_{i,t}^B$ is the real annual bank equity return.

Bank equity crisis. For the new chronology, this is a cumulative 30% decline in bank equity from the previous peak *plus* narrative evidence of widespread bank failures.

Panic banking crisis. A crisis episode identified from the panic database.

BVX Crisis List. The union of bank equity crises and panic banking crises. Crisis start dates are set as the year when bank equity first falls more than 30% from the previous peak; when no such cumulative decline exists, the narrative date is used.

Interpretation. This construction yields a chronology that is neither purely market-based nor purely narrative-based. It uses market data to screen for distress and narratives to verify whether the distress looks like a genuine banking crisis.

2.6 Descriptive facts

Several descriptive facts already point toward the paper’s conclusion:

- The annual data set contains nearly 2,500 country-years.
- Among previously identified narrative banking crises, 32.8% do *not* feature panics.
- Nonpanic banking distress becomes much more common in the postwar era, especially after Bretton Woods.
- The new chronology identifies 27 additional crises not contained in prior narrative chronologies.

The authors also emphasize that the bank equity index is most informative about large commercial banks. This is a limitation for crises centered on small regional banks, private banks, or nonbank financial institutions.

Interpretation. Even before the regressions, the raw historical evidence suggests that a panic-only definition of crisis is too narrow.

3 Model and empirical specifications (all equations + interpretation)

3.1 Event-time comparison around bank equity crashes

The paper first studies simple event-time averages around bank equity crashes. A crash year is defined as $t = 0$, where the bank equity index falls by more than 30% in year t . Outcomes are then traced from $t = -5$ to $t = +5$.

The two key descriptive outcomes are:

- cumulative real GDP growth relative to $t = -5$,
- the change in bank credit-to-GDP relative to $t = -5$.

Interpretation. This is not a causal design. It is a visual way to show that bank equity crashes are preceded by credit booms and followed by persistent weakness in output and bank lending.

3.2 Flexible local projection specification

The core empirical design is a Jorda-style local projection. For horizon $h = 1, \dots, 6$, the paper estimates

$$Y_{i,t+h}^{(h)} = \alpha_i^h + \sum_j \beta_j^h \mathbf{1}\{r_{i,t}^B \in B_j\} + \sum_j \delta_j^h \mathbf{1}\{r_{i,t}^N \in B_j\} + \Gamma^h X_{i,t} + \varepsilon_{i,t}^h,$$

where:

- $Y_{i,t+h}^{(h)}$ is either cumulative real GDP growth from t to $t+h$ or the change in bank credit-to-GDP from t to $t+h$,
- $r_{i,t}^B$ is the bank equity return,
- $r_{i,t}^N$ is the nonfinancial equity return,
- B_j are eight return bins:
 $(-\infty, -45\%], (-45\%, -30\%], (-30\%, -15\%], (-15\%, 0\%], (0\%, 15\%], (15\%, 30\%], (30\%, 45\%], (45\%, \infty)$,

with (0%, 15%] omitted,

- $X_{i,t}$ includes contemporaneous and three lags of GDP growth and the change in credit-to-GDP, together with lags of the bank and nonfinancial return bins.

Country fixed effects are always included. Some robustness checks add year fixed effects. Standard errors are double-clustered by country and year.

Interpretation. This specification asks how future output and bank lending evolve after different sizes of bank equity moves, while stripping out what is already visible in nonfinancial stock returns and recent macro conditions.

3.3 Crash-indicator specification

For exposition, the paper also estimates a simpler crash specification:

$$Y_{i,t+h}^{(h)} = \alpha_i^h + \gamma_t^h + \beta^h \text{Crash}_{i,t}^B + \delta^h \text{Crash}_{i,t}^N + \Gamma^h X_{i,t} + \varepsilon_{i,t}^h,$$

where

$$\text{Crash}_{i,t}^B = \mathbf{1}\{r_{i,t}^B \leq -30\%\}, \quad \text{Crash}_{i,t}^N = \mathbf{1}\{r_{i,t}^N \leq -30\%\}.$$

Interpretation. This compresses the nonlinear specification into the single, easy-to-read question: what happens after a large bank equity crash, relative to normal times?

3.4 Panic interaction specification

To separate crises with and without panics, the authors estimate:

$$Y_{i,t+h}^{(h)} = \alpha_i^h + \beta_1^h \text{Crash}_{i,t}^B + \beta_2^h \text{Panic}_{i,t} + \beta_3^h (\text{Crash}_{i,t}^B \times \text{Panic}_{i,t}) + \Gamma^h X_{i,t} + \varepsilon_{i,t}^h.$$

Interpretation of the coefficients:

- β_1^h : nonpanic bank equity crash episodes,
- β_2^h : panic episodes without a bank equity crash,
- $\beta_1^h + \beta_2^h + \beta_3^h$: panic episodes that *also* have a bank equity crash.

The paper also estimates an even stricter version where the crash indicator is redefined to require narrative evidence of widespread bank failures.

Interpretation. This is the specification that directly speaks to the paper's headline question. If β_1^h is economically large, then panics are not necessary for banking distress to damage the economy.

3.5 Monthly timing analysis

Using monthly data around BVX crises, the paper compares the timing of:

- the first 30% cumulative decline in bank equity from the prior peak,
- the first month of a panic,
- narrative crisis start dates,
- spikes in bank and corporate credit spreads,

- large nonfinancial equity declines.

Interpretation. This section does not estimate a structural model. It asks which indicator moves first in real time. If bank equity leads panics and spread spikes, then market participants are recognizing bank losses before panic-like funding disruptions become visible.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main coefficients?

There is **no external instrument** in the paper. Identification comes from within-country variation over time in bank equity declines, conditional on:

- country fixed effects,
- current and lagged GDP growth,
- current and lagged changes in bank credit-to-GDP,
- nonfinancial equity returns, and
- lags of bank and nonfinancial return bins.

Interpretation. The paper does not claim to isolate an exogenous banking shock. Instead, it shows that *bank-specific market distress* contains powerful incremental information about future macro outcomes.

4.2 Why controlling for nonfinancial equity matters

A fall in bank equity could reflect either:

- a general macro downturn visible to all stocks, or
- losses specific to the banking sector.

By controlling for nonfinancial equity returns, the paper isolates the predictive content of bank equity *relative* to general stock market movements.

Interpretation. This is crucial. If bank equity crashes still predict future GDP and credit after conditioning on nonfinancial equity, the information in bank stocks is not just a re-labeling of a broader recession signal.

4.3 Why bank equity is a plausible crisis signal

Bank equity is an especially informative signal because:

- equity holders are the first-loss claimants,
- equity is sensitive to losses whether banks are near or far from insolvency,
- debt-like funding instruments are often information-insensitive until a panic begins.

Interpretation. This is the conceptual reason the paper can detect “quiet” crises: equity prices respond to expected bank losses before depositors or wholesale creditors necessarily run.

4.4 Evidence that the market measure is not pure noise

The authors show that large bank equity declines are associated with other classic signs of banking crises:

- widespread bank failures,
- high nonperforming loans,
- government intervention in the banking sector,
- large future contractions in bank credit.

They also find that the nonpanic results survive a stricter definition that requires narrative evidence of widespread bank failures, and they survive even when observations near panics and failures are excluded (the “quiet crisis” exercises).

Interpretation. These cross-checks matter because raw equity crashes could otherwise be dismissed as stock-market sentiment or noise.

4.5 What the panic regressions do *not* identify

The paper is explicit that the panic interaction design is **not** a causal estimate of the effect of panics. Panic episodes may be selected on severity, and panic classifications are narrative objects rather than randomized treatments.

Interpretation. The correct reading is:

- panic episodes are usually *worse*,
- but nonpanic distress is still *bad enough* to matter a lot,
- and the fact that panic dates come after equity losses suggests panics are typically an amplification mechanism.

4.6 Limitations

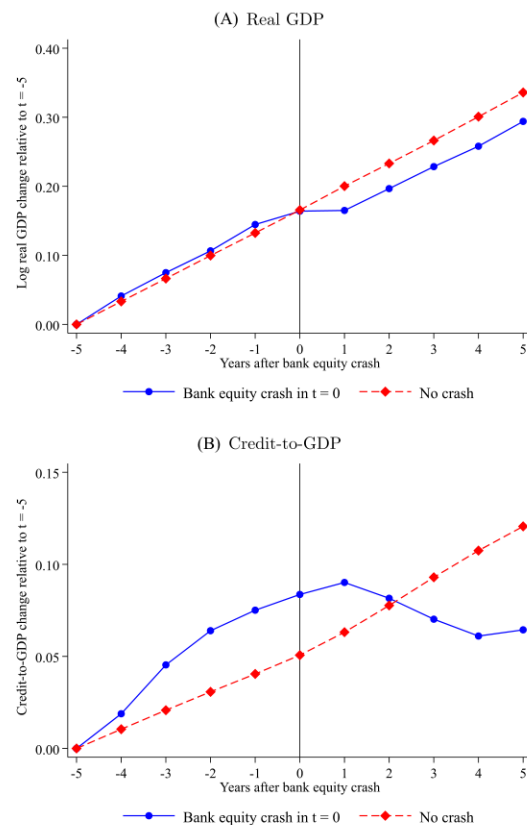
The main limitations are:

- the bank equity index mainly covers large public commercial banks, so regional-bank or nonbank crises can be undermeasured;
- annual returns can miss very short-lived within-year stress episodes;
- the 30% crash threshold is a practical screen, not a theorem;
- some apparent panics without bank equity crashes are likely due to measurement error or near misses rather than true “sunspot” panics.

Interpretation. These limitations mostly work against the paper finding strong effects of nonpanic distress. If anything, they make the paper’s headline result more persuasive.

5 Tables and figures

5.1 Figure I: Dynamics of output and credit around bank equity crashes



Read.

- Real GDP behaves similarly in crash and noncrash years before $t = 0$, but then falls behind sharply after the crash.
- Bank credit-to-GDP rises strongly in the years leading up to a crash, consistent with a precrisis credit boom.
- After the crash, credit stops expanding and then contracts relative to the no-crash benchmark.

Economic message. This figure gives the paper's basic macro narrative in one picture: banking crises grow out of credit booms, and bank equity crashes are followed by persistent output gaps and weaker bank lending.

5.2 Figure II: Bank equity crashes predict output gaps and credit contractions

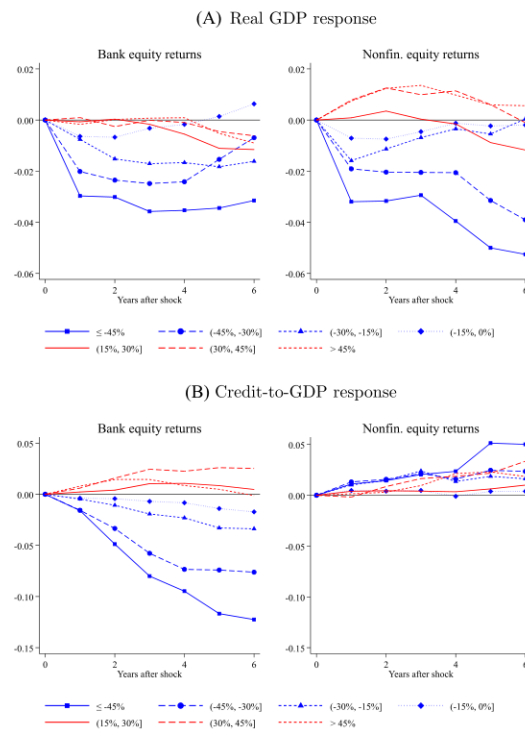


FIGURE II
Bank Equity Crashes Predict Output Gaps and Credit Contractions

Read.

- Large negative bank equity returns predict large subsequent GDP losses.
- Large negative bank equity returns also predict strong contractions in bank credit-to-GDP.
- The response is highly nonlinear: large negative shocks matter much more than moderate or positive shocks.
- Nonfinancial equity declines predict future GDP, but they do *not* predict the same bank credit contraction.

Economic message. Bank equity is not merely another recession indicator. It contains banking-sector information specifically tied to future weakness in intermediated credit.

5.3 Table II: Impact of banking distress with and without panics

TABLE II
IMPACT OF BANKING DISTRESS WITH AND WITHOUT PANICS

Panel A: Baseline

	Real GDP growth _{<i>i,t+3</i>}			Credit-GDP change _{<i>i,t+3</i>}		
	(1)	(2)	(3)	(4)	(5)	(6)
Bank equity crash	-0.030*** [-3.05]	-0.027*** [-3.57]	-0.023*** [-2.88]	-0.051*** [-3.36]	-0.035** [-2.67]	-0.032*** [-2.71]
Panic	-0.017 [-1.05]	0.0018 [0.13]	0.017 [1.68]	-0.018 [-0.96]	0.00980 [0.041]	0.0014 [0.069]
Bank equity crash $\times$ Panic	-0.025 [-1.44]	-0.021 [-1.41]	-0.034** [-2.64]	-0.053* [-1.74]	-0.055* [-1.69]	-0.051 [-1.62]
Nonfinancial equity crash	-0.030** [-2.60]	-0.023*** [-2.85]	-0.024** [-2.36]	0.0098 [0.86]	0.0036 [0.27]	-0.0045 [-0.33]
Country fixed effects	✓	✓	✓	✓	✓	✓
Controls		✓	✓		✓	✓
Year fixed effects			✓			✓
Adj. R^2 (within)	0.05	0.11	0.07	0.03	0.15	0.13
N	2,548	2,548	2,548	2,536	2,536	2,536

Read.

- In the baseline controlled specification, a nonpanic bank equity crash predicts about 2.7% lower real GDP and 3.5 percentage points lower bank credit-to-GDP after three years.
- The panic interaction term is negative, especially for bank credit, so episodes with both a panic and a bank equity crash are substantially worse.
- A panic *without* a bank equity crash has a coefficient close to zero.

Economic message. The crisis does not begin with panic alone. The economically important episodes are the ones with large bank losses, with panics acting mainly as an additional amplifier.

5.4 Figure III: Banking distress with and without banking panics

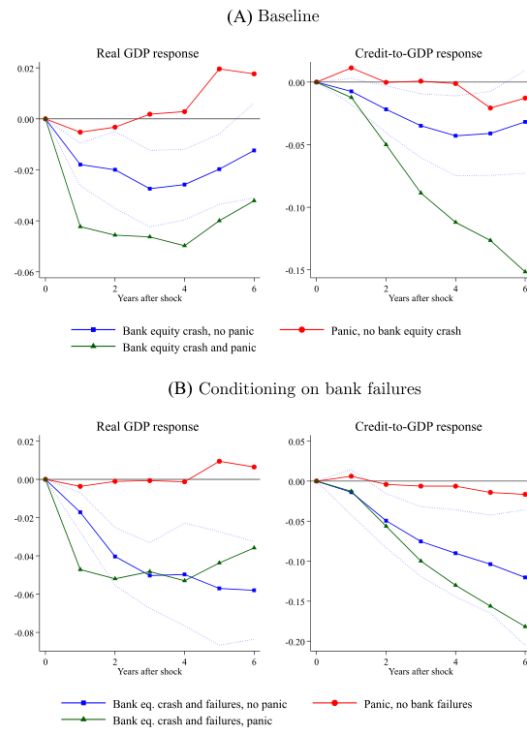


FIGURE III
Banking Distress with and without Banking Panics

Read.

- The blue line shows that nonpanic bank equity crashes still generate meaningful output gaps and credit contraction.
- The green line shows that bank equity crashes *with* panics are even more severe.
- The red line shows that panic episodes without bank equity crashes have little macro effect.
- In Panel B, once crashes are conditioned on widespread bank failures, nonpanic distress is nearly as severe as panic distress.

Economic message. Panics matter, but the paper's main lesson survives the strictest test: large bank losses can damage the economy profoundly even when no panic occurs.

5.5 Table III: Timing of bank equity crashes relative to panics and credit spread spikes

TABLE III
TIMING OF BANK EQUITY CRASHES RELATIVE TO PANICS, CREDIT SPREAD SPIKES, AND NONFINANCIAL EQUITY CRASHES

Panel A: Bank equity crashes detect the crisis before panics, narrative crisis dates, and credit spread spikes

	Before panic	Before Reinhart-Rogoff start dates	Before earliest narrative start dates	Before 2% spike in bank credit spread	Before 1% spike in bank credit spread	Before 2% spike in corp credit spread	Before 1% spike in corp credit spread
Average (in months, signed)	7.46***	3.20**	2.85**	6.18***	3.44**	8.84***	4.26*
<i>t</i> -stat	4.92	2.48	2.34	6.14	2.03	6.75	1.80
<i>N</i>	93	93	101	40	41	19	19
Pos.	69	38	32	32	23	16	12
Zero	5	33	53	4	2	1	0
Neg.	19	22	16	4	16	2	7
$\frac{Pos.}{Pos.+Neg.}$	78.4%***	63.3%**	66.6%**	88.9%***	58.9%	88.9%***	63.2%
<i>p</i> -value	.000	.026	.015	.000	.168	.001	.180

Read.

- The bank equity crash occurs on average 7.46 months before the panic.
- It also precedes the earliest narrative crisis start date by 2.85 months and the Reinhart–Rogoff start date by 3.20 months.
- Bank equity crashes arrive well before large credit spread spikes: 6.18 months before a 2% spike in bank credit spreads and 8.84 months before a 2% spike in corporate credit spreads.

Economic message. Bank equity is an early-warning indicator. Market participants price in bank losses before panic runs and spread spikes make the crisis visible in debt markets.

6 Short summary

- The paper builds a new 46-country historical data set on bank equity returns, nonfinancial returns, panics, and bank failures from 1870 to 2016.
- Large bank equity declines strongly predict later real GDP losses and contractions in bank credit-to-GDP, even after controlling for nonfinancial equity returns.
- Panics amplify crises, but nonpanic banking distress is also associated with deep recessions and persistent credit weakness.
- Bank equity losses usually come *before* panics and credit spread spikes.
- A new crisis chronology based on bank equity losses plus narrative evidence uncovers many omitted “quiet” crises and sharpens the historical classification of banking crises.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that banking crises are not synonymous with banking panics. Severe bank distress can translate into output gaps and credit contraction even in the absence of a creditor run, and large bank equity declines are a powerful way to detect such episodes.

7.2 Economic intuition

Bank equity reflects expected and realized losses on bank balance sheets. Once those losses erode capital, banks cut back on lending, and the economy weakens through a bank capital crunch. Sometimes creditor runs follow and make the downturn worse, but the underlying damage is often already there.

7.3 Takeaway

For real-time monitoring and for historical chronology, regulators and researchers should track bank equity alongside credit growth and funding spreads. A panic-only view misses a large class of economically important “quiet” banking crises.